

RESEARCH ARTICLE

Brain morphometry patterns in the presence of Alzheimer's disease and/or LATE neuropathologic changes

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Abstract

INTRODUCTION: Alzheimer's disease neuropathologic change (ADNC) and limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy neuropathologic change (LATENC) are common in older adults, yet differences in brain morphometry patterns when one or both pathologies are present remain unclear.

METHODS: We used deformation-based morphometry on ex-vivo MRI from 912 community-based older adults to compare groups with or without ADNC and/or LATENC.

RESULTS: AD+LATE- and AD-LATE+ groups showed less tissue in the medial temporal lobe than AD-LATE-. The AD+LATE+ group had less tissue in temporal, frontal, and parietal lobes. The AD-LATE+ group exhibited smaller anterior hippocampi than the AD+LATE- group. These findings were less pronounced in individuals without dementia. Increments in LATENC stages were associated with smaller hippocampi than increments in ADNC severity, independent of the severity of comorbid ADNC or LATENC, respectively.

DISCUSSION: These findings reveal distinct and overlapping brain morphometry patterns associated with ADNC and/or LATENC, with possible implications for diagnosis in older adults.

KEYWORDS

Alzheimer's disease neuropathologic change (ADNC), deformation-based morphometry (DBM), limbic-predominant age-related TDP-43 encephalopathy neuropathologic change (LATENC), magnetic resonance imaging (MRI), neurodegeneration, neuropathology

Highlights

- Alzheimer's disease neuropathologic change (ADNC) and limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43)

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encephalopathy neuropathologic change (LATENC) have distinct and overlapping brain morphometry patterns

- ADNC and LATENC were associated with less tissue mainly in the medial temporal lobe
- Mixed ADNC and LATENC showed less tissue in more regions than either alone
- LATENC showed a smaller anterior hippocampus than ADNC, even without dementia
- Higher LATENC stages showed smaller hippocampi than more severe ADNC

1 | BACKGROUND

Age-related cognitive decline is now recognized as the result of multiple, possibly comorbid pathologies rather than explained by a single disease. Although Alzheimer's disease neuropathologic change (ADNC) remains a key contributor to dementia, it rarely acts alone and does not fully account for the heterogeneity of late-life cognitive impairment. Postmortem studies have revealed that mixed pathologies are common in older adults, often encompassing both neurodegenerative and vascular pathologies.¹ Among these, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy neuropathologic change (LATENC) has emerged as a frequent and distinct contributor to neurodegeneration and cognitive decline in late life.² LATENC is reported in 25%–40% of individuals older than 85 years of age in large community-based autopsy series, even in the absence of substantial ADNC, appearing in one in four brains with no or minimal ADNC, and in about half of severe ADNC cases. TDP-43 pathology, the hallmark of LATENC, occasionally colocalizes with tau-immunoreactive neurofibrillary tangles.^{4–6} Because clinical outcomes vary depending on the combination of pathologies,⁷ a nuanced understanding of the distinct and combined effects of ADNC and LATENC on the aging brain is important for accurate diagnosis and biomarker development.

Both LATENC and ADNC lead to medial temporal lobe atrophy and episodic memory decline, making them difficult to distinguish clinically and through standard imaging.^{8–10} Recent studies combining data from imaging and neuropathology have revealed that LATENC exhibits an atrophy pattern involving primarily the anterior hippocampus, entorhinal cortex, amygdala, and orbitofrontal regions, closely paralleling the distribution of TDP-43 pathology.^{2,11–15} In contrast, ADNC tends to show more posterior hippocampal atrophy,^{15–19} with anterior-to-posterior atrophy ratios showing relatively good discrimination between ADNC cases with and without LATENC.¹⁵ These findings demonstrate that despite the large overlap in clinical symptoms and imaging features, LATENC and ADNC also have diverging macrostructural signatures that may hold diagnostic value.

Mixed ADNC and LATENC is associated with worse clinical outcomes than either pathology alone. Individuals with comorbid ADNC and LATENC exhibit faster hippocampal atrophy, greater memory

impairment, and more rapid global cognitive decline.^{20–23} Although pure LATENC exhibits a slow progression of clinical symptoms, its presence alongside ADNC accelerates neurodegeneration and lowers the threshold for dementia.^{8,24–26} These findings underscore the importance of identifying and understanding mixed ADNC and LATENC in the aging brain.

Many imaging studies have focused on ADNC and LATENC separately,^{27–29} yet few have examined their combined effects on the aging brain.^{24,30} Among the few that did, analyses were largely focused on cognition^{9,24} and regional volumetry,³¹ primarily in the medial temporal lobe^{15,17,32} and the hippocampus,^{11,13,30,33} rather than the whole-brain. Therefore, a systematic evaluation of the association of ADNC and LATENC with brain macrostructure while accounting for other common comorbid pathologies is warranted.

This study combined voxel-wise deformation-based morphometry (DBM)^{34,35} on ex vivo magnetic resonance imaging (MRI) and detailed neuropathologic findings in a large number of community-based older adults that came to autopsy to delineate and compare the brain morphometry patterns of ADNC, LATENC, and mixed ADNC and LATENC. The investigation was repeated in a subset of older adults without dementia. By controlling for other age-related neuropathologies and demographics, this study generated statistically independent brain morphometry patterns, offering critical insights into the macrostructural signatures of ADNC, LATENC, and their co-occurrence, with potential implications for diagnosis in aging populations.

2 | METHODS

2.1 | Participants and data collection

This study analyzed cerebral hemispheres from older adults enrolled in one of four ongoing longitudinal, clinical-pathologic cohort studies of aging: the Rush Memory and Aging Project (MAP), the Minority Aging Research Study (MARS),³⁶ the Religious Orders Study (ROS)³⁷ and the African American Clinical Core (AA Core) of the Rush Alzheimer's Disease Research Center (Rush ADRC).³⁸ Study procedures are harmonized across the cohorts, and all evaluations are conducted by the same research team to facilitate valid comparisons.

RESEARCH IN CONTEXT

1. **Systematic review:** The authors reviewed existing literature on magnetic resonance imaging (MRI)–pathology studies investigating the individual and combined effects of Alzheimer's disease neuropathologic change (ADNC) and limbic-predominant age-related TDP-43 encephalopathy neuropathologic change (LATENC). Although prior studies link each pathology to medial temporal lobe atrophy, few have systematically compared whole-brain morphometry across isolated versus mixed pathologies while adjusting for common comorbidities in a large community-based cohort.
2. **Interpretation:** Our findings demonstrate that although ADNC and LATENC exhibit overlapping morphometry patterns in the medial temporal lobe, LATENC is associated with less tissue in the anterior hippocampus, even without dementia. The co-occurrence of the two pathologies is associated with less tissue in more brain regions than either pathology alone, suggesting a potential synergistic effect.
3. **Future directions:** Future research should investigate the pathophysiological mechanisms underlying the hypothesized synergistic effects of ADNC and LATENC. In addition, the higher sensitivity of the anterior hippocampus to LATENC than to ADNC warrants further investigation.

All four cohort studies were approved by an institutional review board (IRB) at Rush University Medical Center, and all participants consented to annual clinical assessments, signed an anatomic gift act, and agreed to brain donation at the time of death. Annual standardized clinical evaluations, comprehensive neuropsychological testing, medical history reviews, and neurological examinations were performed for all participants. The diagnosis of Alzheimer's dementia followed the criteria of the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association.³⁹ Participants exhibiting cognitive impairment but not meeting the criteria for dementia were classified as having mild cognitive impairment^{40,41} (MCI). Those without dementia or MCI were categorized as having no cognitive impairment⁴² (NCI). At the time of analysis, a total of 4022 individuals from the parent studies had completed baseline clinical evaluation after the ex vivo MRI data collection began. Of these participants, 1746 passed away, 1449 underwent autopsy, and 1202 had ex vivo MRI and neuropathology data. Of these, the first 922 consecutive participants had undergone ex vivo MRI image processing and quality checks at the time of analysis, and 10 individuals were excluded due to large brain abnormalities that precluded reliable image registration and morphometric analysis (e.g., very large infarcts, tumors). Therefore, the final analyses were performed on the remaining 912 participants.

2.2 | Ex vivo MRI data acquisition and processing

At autopsy, a technician extracted the brain and divided it into left and right hemispheres. The hemisphere exhibiting more visible pathology was immersed in a phosphate-buffered 4% formaldehyde solution and refrigerated at 4°C within 30 min, whereas the contralateral hemisphere was frozen and stored. In most cases, pathology was not visible from the outside, and therefore the selection of hemisphere was made randomly. Before ex vivo MRI, the refrigerated hemisphere was brought to room temperature and imaged while immersed in formaldehyde, with its medial aspect facing downward. Gross examination was performed within 2 weeks post-MRI, followed by detailed histopathologic examination by a board-certified neuropathologist.³¹

Ex vivo MRI data were acquired on 3T scanners ≈30 days post-mortem using two-dimensional (2D) multi-echo spin-echo (ME-SE) sequences. Due to scanner upgrades over the course of data collection, imaging was conducted on four different scanners. ME-SE acquisitions had a voxel size of $0.6 \times 0.6 \times 1.5 \text{ mm}^3$, and echo times (TEs) that covered at least the range of ≈10–50 ms. To ensure consistency across scanners, only T2-weighted images with TEs between 49.5 and 55 ms were analyzed. These ex vivo T2-weighted images have previously generated volumetric measurements that are linearly related to those made in vivo,⁴³ supporting their appropriateness for morphometric analyses.

T2-weighted images from different participants were intensity-normalized for standardization and non-linearly registered to a brain hemisphere template using Advanced Normalization Tools (ANTs).⁴⁴ The resulting deformation fields were used to compute the log-Jacobian determinant (logJ) at each voxel, capturing localized volume expansion or contraction when compared to the template. The maps of logJ were smoothed using a Gaussian filter with a full-width at half maximum (FWHM) of 4 mm and were used in DBM analyses.

2.3 | Neuropathologic examination

Detailed neuropathologic examination was conducted by a board-certified neuropathologist who was blinded to all imaging and clinical data. Following postmortem MRI, each hemisphere was sectioned coronally into 1-cm thick slabs for macroscopic evaluation. Tissue blocks were selected, embedded in paraffin, cut into 6-μm sections, and stained with hematoxylin and eosin for histological examination.

Pathologic diagnosis of Alzheimer's disease (AD) was determined using the National Institute on Aging–Alzheimer's Association (NIA-AA) criteria⁴⁵ and was rated as no, low, intermediate, or high. Amyloid beta (Aβ) deposits were assessed in seven brain regions (hippocampus, middle temporal, midfrontal, inferior parietal, basal ganglia, substantia nigra, and cerebellum) and described by Thal phases. Staging of neurofibrillary tangles (NFTs) (in hippocampus, entorhinal, midfrontal, inferior parietal, middle temporal cortices) was modified from Braak and Braak. Scoring of neuritic plaques (in midfrontal, inferior parietal, middle temporal cortices) followed the criteria of the Consortium to Establish a Registry for AD (CERAD).

TDP-43 was identified using immunohistochemistry with a phosphorylated TDP-43 monoclonal antibody (TAR5P-1D3, pS409/410; 1:100, Ascenion, Munich, Germany) to detect cytoplasmic TDP-43 inclusions in neurons and glia across eight brain regions: the amygdala, entorhinal cortex, hippocampal CA1 and subiculum, dentate gyrus, temporal pole, middle temporal cortex, orbitofrontal cortex, and middle frontal cortex.⁴⁶ LATENC severity was classified into four stages: stage 0 (absence of TDP-43 inclusions), stage 1 (TDP-43 inclusions restricted to the amygdala), stage 2 (involvement of the amygdala and either the entorhinal cortex or hippocampus), and stage 3 (presence of TDP-43 inclusions in the amygdala, entorhinal cortex or hippocampus, and neocortex).^{2,46}

Lewy bodies were evaluated in seven brain regions: the entorhinal, anterior cingulate, midfrontal, superior or middle temporal, and inferior parietal cortices, as well as the substantia nigra and amygdala. Their presence was categorized as either absent or present.⁴⁷ For each participant, the number of gross infarcts of any age was classified into three categories: none, a single infarct, or two or more infarcts. Microscopic infarcts were identified through histological examination of at least nine brain regions: the midfrontal, middle temporal, entorhinal, hippocampal, inferior parietal, and anterior cingulate cortices, as well as the anterior basal ganglia, thalamus, and midbrain. For each participant, microscopic infarcts were recorded as absent, single, or multiple.⁴⁸ Atherosclerosis severity was assessed at the circle of Willis and classified into four categories: none, mild, moderate, or severe.⁴⁹ Arteriolosclerosis was evaluated in the anterior basal ganglia based on the severity of wall thickening and luminal occlusion of the small arterioles and was rated as none, mild, moderate, or severe. Cerebral amyloid angiopathy (CAA) was examined in four brain regions: the angular, midfrontal, middle temporal, and calcarine cortices. CAA was scored as none, mild, moderate, or severe.⁵⁰

2.4 | Statistical analysis

In this study, we categorized individuals with intermediate or high ADNC as ADNC positive, abbreviated as AD+, and those with no or low ADNC as ADNC negative, abbreviated as AD-. Similarly, individuals with LATENC Stage 2 or 3 were classified as LATENC positive, abbreviated as LATE+, and those with LATENC Stage 0 or 1 were classified as LATENC negative, abbreviated as LATE-. Although there is no direct correspondence between ADNC severity levels and LATENC stages, the selected threshold for ADNC and LATENC positivity is known to be the point below which ADNC severity and LATENC stage are not considered adequate explanations for cognitive impairment and dementia.^{9,51} Demographic, clinical, and neuropathologic characteristics were compared across the AD-LATE-, AD+LATE-, AD-LATE+, and AD+LATE+ groups. Continuous variables were compared using one-way analysis of variance (ANOVA) for normally distributed data and the Kruskal-Wallis test for non-normally distributed data. Categorical variables were analyzed using the chi-square test or Fisher's exact test. Post hoc pairwise comparisons were performed using Tukey's honestly significant difference (HSD) (following ANOVA) or

Mann-Whitney tests (following Kruskal-Wallis) for continuous variables, and chi-square or Fisher's exact tests for categorical variables.

First, we tested the hypothesis that AD+LATE- individuals exhibited lower logJ values, indicative of less tissue, compared to those without either pathology (AD-LATE-). Next, we tested the hypothesis that AD-LATE+ individuals had lower logJ values relative to the AD-LATE- group. In addition, we tested the hypothesis that those that were both AD+ and LATE+ (AD+LATE+) had lower logJ values compared to the AD-LATE- group, in more brain areas than the patterns for AD+LATE- and AD-LATE+. Finally, we tested the hypothesis that AD-LATE+ individuals exhibited lower logJ values than those that were AD+LATE-. To test these hypotheses, voxel-wise linear regression was conducted with local brain deformations (captured by smoothed logJ maps) as the dependent variable and neuropathology group as the independent variable, controlling for other pathologies (Lewy bodies, gross and microscopic infarcts, atherosclerosis, arteriolosclerosis, CAA), age at death, sex, years of education,^{52,53} postmortem interval to immersion in fixative, postmortem interval to imaging, and MRI scanner. Hippocampal sclerosis was not included as a covariate because of its strong association with advanced LATENC and the potential for over-adjustment. All analyses were repeated in a subset of participants without dementia to assess whether the derived spatial patterns were present, and to what extent, in the absence of dementia. Even though the interval between the final cognitive evaluation and death was generally short for participants without dementia (median = 0.79 years), we repeated the analyses after excluding individuals whose final evaluation occurred more than 2 years before death. All analyses were conducted using Permutation Analysis of Linear Models (PALM) in the Functional Magnetic Resonance Imaging of the Brain (FMRIB) Software Library (FSL).⁵⁴ To account for differences in acquisition protocols, variance was modeled separately for each scanner by incorporating four exchangeability blocks—one per scanner—so that permutations were restricted to participants imaged on the same scanner only.⁵⁵ Statistical significance was determined using 5000 permutations, applying Family-Wise Error Rate (FWER) correction to achieve an adjusted threshold of $p < 0.05$. Significant clusters were identified using the Threshold-Free Cluster Enhancement (TFCE) method.

To separate the statistically independent effects of different ADNC severity levels and LATENC stages on hippocampal size, participants were divided into 16 distinct subgroups representing all possible combinations of ADNC severity levels (i) and LATENC stages (j), where $i, j \in \{0, 1, 2, 3\}$, and the median logJ value inside the hippocampus was extracted for each participant. A multivariable ordinary least squares (OLS) regression model was employed to assess differences in hippocampal median logJ values across the 16 subgroups, adjusting for other pathologies, age at death, sex, education, postmortem intervals, and scanner. The p -values were adjusted using Dunnett's method to control for the FWER at $p < 0.05$. In addition, to isolate the incremental effect of ADNC severity levels or LATENC stages, we employed stratified, covariate-adjusted linear models. Within each ADNC severity level, the effect of LATENC stage was evaluated using a partial F-test ($df = 3$) comparing the four stages; an analogous procedure assessed ADNC levels within each LATENC stage. Effect sizes were

TABLE 1 Demographic and clinical characteristics of the participants.

Characteristic	AD-LATE-	AD+LATE-	AD-LATE+	AD+LATE+	All participants
N	218	343	79	272	912
Age at death, years, mean (SD)	89.0 (7.2)	90.6 (6.0)	92.4 (5.7)	92.3 (5.6)	90.6 (6.4)
Male, n (%)	74 (33.9)	100 (29.2)	19 (24.1)	63 (23.2)	256 (28.0)
Education, years, mean (SD)	16.1 (3.3)	15.6 (3.8)	15.5 (3.4)	15.8 (3.4)	15.7 (3.5)
Right hemisphere, n (%)	116 (53.2)	157 (45.8)	36 (45.6)	108 (39.6)	417 (45.7)
Race, n (%)	191 (87.6)	305 (88.9)	74 (93.6)	253 (93.0)	823 (90.2)
– Non-Latino White	21 (9.6)	22 (6.4)	3 (3.8)	15 (5.5)	61 (6.7)
– Non-Latino Black	6 (2.8)	15 (4.4)	1 (1.3)	4 (1.5)	26 (2.9)
– Latino	0 (0)	1 (0.3)	1 (1.3)	0 (0)	2 (0.2)
– Other/Unknown					
Clinical diagnosis, n (%)	130 (59.6)	106 (30.9)	27 (34.2)	30 (11.0)	293 (32.1)
– Mild cognitive impairment	54 (24.8)	80 (23.3)	21 (26.6)	53 (19.5)	208 (22.8)
– Dementia	34 (15.6)	157 (45.8)	31 (39.2)	189 (69.5)	411 (45.1)
At least one copy of the APOE ε4 allele, n (%)	32 (14.7)	89 (25.9)	14 (17.7)	109 (40.1)	244 (26.8)
Mini-Mental State Examination ^a (MMSE), mean (SD)	25.1 (6.4)	20.1 (9.2)	21.7 (8.2)	14.8 (10.1)	19.9 (9.6)
Postmortem interval to fixation, hours, mean (SD)	10.2 (7.0)	10.2 (8.5)	9.2 (7.7)	9.1 (6.4)	9.8 (7.5)
Postmortem interval to imaging, days, mean (SD)	38.1 (23.3)	35.2 (11.8)	35.1 (11.9)	37.2 (15.2)	36.5 (16.3)
MRI Scanner, n (%)	19 (8.7)	22 (6.4)	5 (6.3)	29 (10.7)	75 (8.2)
– 3T GE Signa	24 (11.0)	32 (9.3)	6 (7.6)	20 (7.3)	82 (9.0)
– 3T Siemens Trio	64 (29.4)	121 (35.3)	18 (22.8)	79 (29.0)	282 (30.9)
– 3T Philips Achieva	111 (50.9)	168 (49.0)	50 (63.3)	144 (53.0)	473 (51.9)
– 3T Siemens Verio					

^aProximate to death.

ADNC, Alzheimer's disease neuropathologic change; APOE, apolipoprotein E; LATENC, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy neuropathologic change; SD, standard deviation.

AD+/AD- and LATE+/LATE- indicate binary neuropathologic group classifications based on ADNC and LATENC, respectively. AD+ denotes intermediate/high ADNC, AD- denotes no/low ADNC, LATE+ denotes LATENC Stage 2/3, and LATE- denotes LATENC Stage 0/1.

quantified using partial R^2 , and stage-wise differences were determined via Dunnett-adjusted estimated marginal mean comparisons against the stage = 0 reference. This approach allowed for an assessment of the impact of each pathology on hippocampal logJ holding the other pathology constant. OLS models were implemented in R (version 4.4.1), and post hoc comparisons via Dunnett's method were performed using the emmeans (version 2.0.1) and multcomp (version 1.4.29) packages.

3 | RESULTS

3.1 | Demographic, clinical, and neuropathologic characteristics

Demographic and clinical characteristics for the participants ($N = 912$) are summarized in Table 1. Participants had a mean age at death of 91 ($SD = 6$) years, 90% were White and 7% were Black. Proximate to death, 32% had no cognitive impairment, 23% had MCI, and 45% had dementia. Neuropathologic findings (Table 2) revealed that 67% of participants exhibited intermediate or high ADNC (defined herein as AD+), whereas 38% presented with LATENC Stage 2 or 3 (defined as LATE+).

Individuals in the LATE+ groups (AD-LATE+ and AD+LATE+) were ≈ 2 –3 years older than those in the LATE- groups (AD+LATE- and AD-LATE-). According to clinical diagnosis and MMSE scores, cognitive impairment was most severe in the AD+LATE+ group, followed by the AD+LATE- and AD-LATE+ groups, which did not differ significantly from one another; the AD-LATE- group exhibited the least impairment. The AD+ groups (AD+LATE- and AD+LATE+) demonstrated a higher frequency of apolipoprotein E (APOE) ε4 carrier status compared to the AD- groups (AD-LATE- and AD-LATE+), with 40% of individuals in the AD+LATE+ group being APOE ε4 carriers. In addition, the AD+ groups were characterized by more severe CAA than the AD- groups.

3.2 | Brain morphometry pattern for the AD+LATE- and AD-LATE+ groups

Voxel-wise linear regression revealed the brain morphometry patterns for the AD+LATE- and AD-LATE+ groups. Compared to AD-LATE- older adults, AD+LATE- individuals had lower logJ values, indicative of less tissue, in the amygdala, entorhinal cortex, hippocampus, parahippocampal gyrus, fusiform gyrus, temporal pole, insula, and medial orbitofrontal gyrus (Figure 1). AD-LATE+ individuals

TABLE 2 Neuropathologic characteristics of the participants.

Characteristic	AD-LATE-	AD+LATE-	AD-LATE+	AD+LATE+	All participants
ADNC (NIA-AA), <i>n</i> (%)	82 (37.6)	-	28 (35.4)	-	110 (12.1)
– No AD neuropathologic change	136 (62.4)	-	51 (64.6)	-	187 (20.5)
– Low AD neuropathologic change	-	235 (68.5)	-	146 (53.7)	381 (41.8)
– Intermediate AD neuropathologic change	-	108 (31.5)	-	126 (46.3)	234 (25.6)
– High AD neuropathologic change	-	-	-	-	-
LATENC, <i>n</i> (%)	164 (75.2)	238 (69.4)	-	-	402 (44.1)
– No TDP-43 inclusions	54 (24.8)	105 (30.6)	-	-	159 (17.4)
– TDP-43 inclusions in amygdala only	-	-	35 (44.3)	68 (25)	103 (11.3)
– TDP-43 inclusions in amygdala and entorhinal cortex or hippocampus CA1	-	-	44 (55.7)	204 (75)	248 (27.2)
– TDP-43 inclusions in amygdala and entorhinal cortex or hippocampus CA1 and neocortex	-	-	-	-	-
Lewy bodies, <i>n</i> (%)	47 (21.6)	97 (28.3)	23 (29.1)	92 (33.8)	259 (28.4)
Gross infarcts, <i>n</i> (%)	91 (41.7)	133 (38.8)	33 (41.8)	112 (41.2)	369 (40.5)
Microscopic infarcts, <i>n</i> (%)	84 (38.5)	132 (38.5)	30 (38.0)	107 (39.3)	353 (38.7)
Atherosclerosis, <i>n</i> (%)	61 (28.0)	82 (23.9)	14 (17.7)	43 (15.8)	200 (21.9)
– None	107 (49.1)	186 (54.2)	43 (54.4)	148 (54.4)	484 (53.1)
– Mild	39 (17.9)	62 (18.1)	18 (22.8)	65 (23.9)	184 (20.2)
– Moderate	11 (5.0)	13 (3.8)	4 (5.1)	16 (5.9)	44 (4.8)
– Severe	-	-	-	-	-
Cerebral amyloid angiopathy, <i>n</i> (%)	95 (43.6)	35 (10.2)	32 (40.5)	24 (8.8)	186 (20.4)
– None	78 (35.8)	165 (48.1)	30 (38.0)	115 (42.3)	388 (42.5)
– Mild	33 (15.1)	100 (29.2)	9 (11.4)	84 (30.9)	226 (24.8)
– Moderate	12 (5.5)	43 (12.5)	8 (10.1)	49 (18.0)	112 (12.3)
– Severe	-	-	-	-	-
Arteriolosclerosis, <i>n</i> (%)	90 (41.3)	119 (34.7)	30 (38.0)	85 (31.3)	324 (35.5)
– None	75 (34.4)	137 (39.9)	29 (36.7)	98 (36.0)	339 (37.2)
– Mild	35 (16.0)	70 (20.4)	17 (21.5)	65 (23.9)	187 (20.5)
– Moderate	18 (8.3)	17 (5.0)	3 (3.8)	24 (8.8)	62 (6.8)
– Severe	-	-	-	-	-

ADNC, Alzheimer's disease neuropathologic change; CA1, cornu ammonis; LATENC, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy neuropathological change; NIA-AA: National Institute on Aging—Alzheimer's Association; SD, standard deviation. AD+/AD- and LATE+/LATE- indicate binary neuropathologic group classifications based on ADNC and LATENC, respectively. AD+ denotes intermediate/high ADNC, AD- denotes no/low ADNC, LATE+ denotes LATENC Stage 2/3, and LATE- denotes LATENC Stage 0/1.

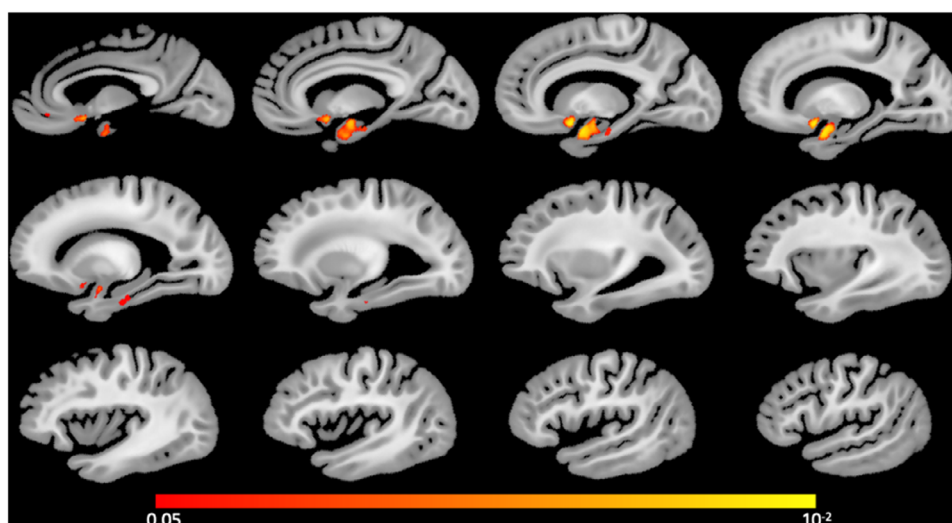


FIGURE 1 Regions with significantly less tissue in the AD+LATE- group compared to the AD-LATE- group controlling for other neuropathologies, demographics, and other covariates. The color-scale represents family-wise error rate (FWER)-corrected *p*-values. AD, Alzheimer's disease; LATE, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy.

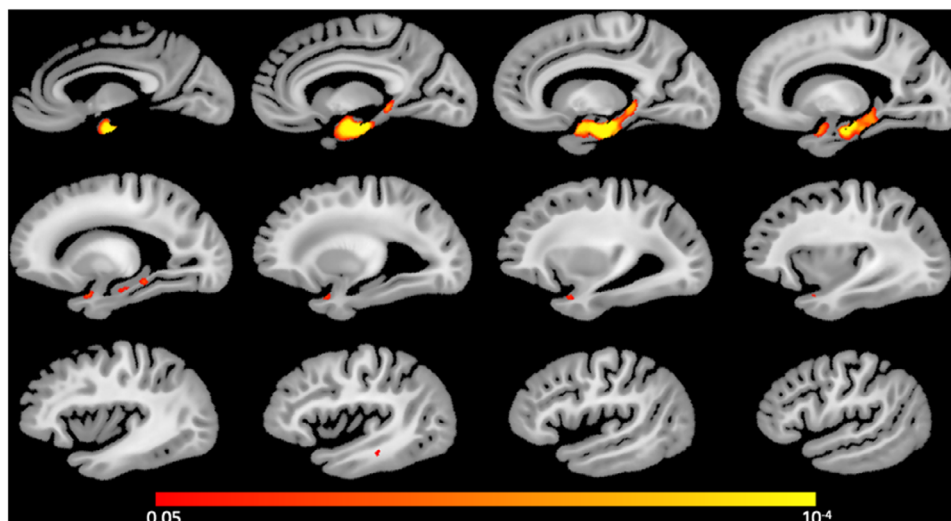


FIGURE 2 Regions with significantly less tissue in the AD-LATE+ group compared to the AD-LATE- group controlling for other neuropathologies, demographics, and other covariates. The color-scale represents family-wise error rate (FWER)-corrected p -values. AD, Alzheimer's disease; LATE, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy.

exhibited less tissue in the amygdala, entorhinal cortex, hippocampus, parahippocampal gyrus, fusiform gyrus, temporal pole, superior temporal gyrus, and isthmus of the cingulate, compared to AD-LATE- older adults (Figure 2). No brain regions showed less tissue in the AD-LATE- group compared to the AD+LATE- or the AD-LATE+ groups. When repeating the same analyses in older adults without dementia, AD+LATE- individuals did not differ from the AD-LATE- group, whereas the AD-LATE+ group showed less tissue in the hippocampus (Figure S1). These findings in individuals without dementia were essentially unchanged after excluding those whose final clinical evaluation occurred more than 2 years before death ($N = 82$).

3.3 | Brain morphometry pattern for the AD+LATE+ group

Compared to AD-LATE- individuals, those that were AD+LATE+ had less tissue in more brain regions than those shown in the morphometry patterns for AD+LATE- and AD-LATE+ older adults (Figure 3). Specifically, the AD+LATE+ morphometry pattern included the amygdala; entorhinal cortex; hippocampus; parahippocampal gyrus; fusiform gyrus; temporal pole; inferior, middle, and superior temporal gyrus; insula; banks of the superior temporal sulcus; medial orbitofrontal gyrus; superior frontal gyrus; rostral middle frontal gyrus; rostral anterior cingulate; inferior and superior parietal gyrus; cuneus; precuneus; and lingual gyrus. No brain regions showed less tissue in the AD-LATE- group compared to the AD+LATE+ group. When repeating the analysis in older adults without dementia, less tissue was again observed in the hippocampus and medial temporal and frontal lobe regions of the AD+LATE+ group compared to the AD-LATE- group (Figure S2). These findings in individuals without dementia were essentially unchanged after excluding those whose final clinical evaluation occurred more than 2 years before death.

3.4 | Morphometric differences between the AD-LATE+ and AD+LATE- groups

Voxel-wise linear regression comparing the AD-LATE+ and AD+LATE- groups showed that AD-LATE+ individuals had less tissue in the anterior hippocampus, also extending into entorhinal and parahippocampal regions, compared to those that were AD+LATE- (Figure 4). No brain regions showed less tissue in the AD+LATE- group compared to the AD-LATE+ group. When repeating the analysis in older adults without dementia, less tissue was again observed in the hippocampus of the AD-LATE+ group (Figure S3). This finding in individuals without dementia was essentially unchanged after excluding those whose final clinical evaluation occurred more than 2 years before death.

3.5 | Hippocampal size for different ADNC severity levels and LATENC stages

Comparison of the 16 subgroups representing all possible combinations of ADNC severity and LATENC stages demonstrated that the least tissue in the hippocampus was found in subgroups with end-stage LATENC (Stage 3) and comorbid ADNC, with the largest negative effects observed in the ADNC1/LATENC3 ($\beta = -0.195$, $p = 1.3 \times 10^{-8}$), ADNC2/LATENC3 ($\beta = -0.154$, $p = 3.5 \times 10^{-10}$), and ADNC3/LATENC3 ($\beta = -0.139$, $p = 1.4 \times 10^{-7}$) subgroups (Figure 5). Less tissue in the hippocampus compared to ADNC0/LATENC0 was also observed in subgroups ADNC3/LATENC2 ($\beta = -0.110$, $p = 5.2 \times 10^{-3}$), ADNC2/LATENC2 ($\beta = -0.088$, $p = 0.049$), and ADNC3/LATENC1 ($\beta = -0.087$, $p = 0.044$), yet with smaller effect sizes than those observed in ADNC1/LATENC3, ADNC2/LATENC3 and ADNC3/LATENC3 subgroups (Figure 5). Notably, subgroups with both ADNC severity 1 or 2 and LATENC stages 1 or 2 (i.e.,

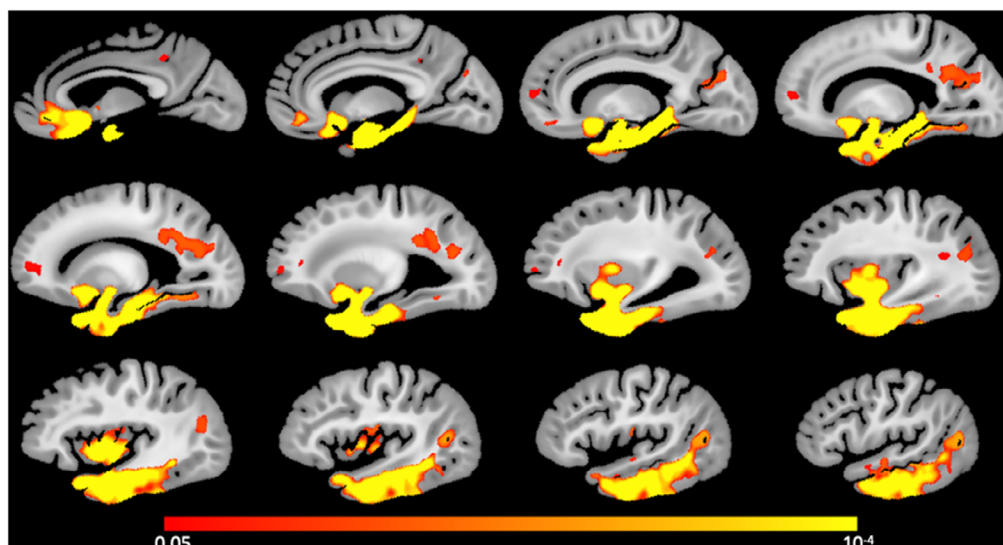


FIGURE 3 Regions with significantly less tissue in the AD+LATE+ group compared to the AD-LATE- group controlling for other neuropathologies, demographics and other covariates. The color-scale represents family-wise error rate (FWER)-corrected p -values. AD, Alzheimer's disease; LATE, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy.

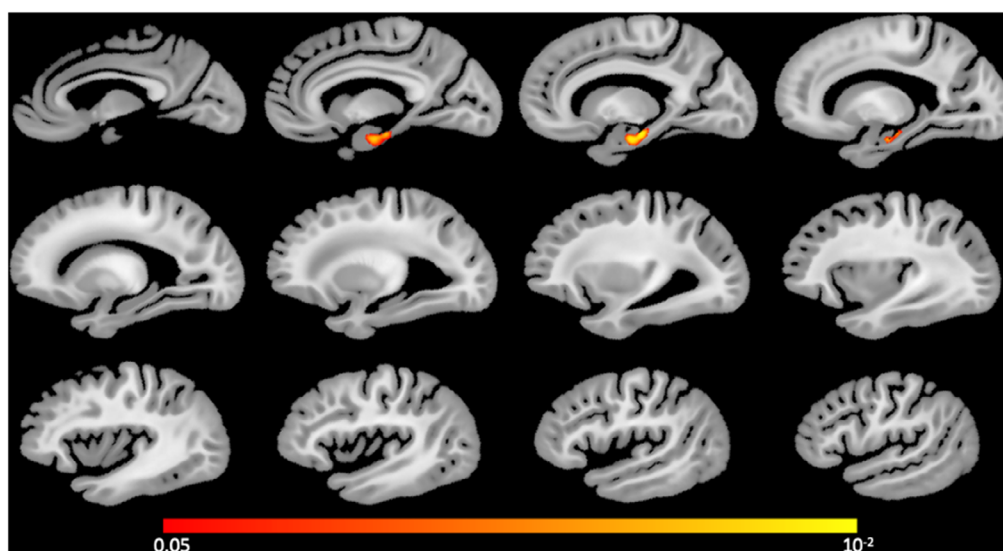


FIGURE 4 Regions with significantly less tissue in the AD-LATE+ group compared to the AD+LATE- group controlling for other neuropathologies, demographics, and other covariates. The color-scale represents family-wise error rate (FWER)-corrected p -values. AD, Alzheimer's disease; LATE, limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy.

subgroups: ADNC1/LATENC1, ADNC1/LATENC2, ADNC2/LATENC1) did not show significant differences in the hippocampus compared to the ADNC0/LATENC0 subgroup ($p > 0.1$) (Figure 5). Furthermore, subgroups with any ADNC severity level but no LATENC (ADNC1-3/LATENC0), and those with any LATENC stage but no ADNC (ADNC0/LATENC1-3) also did not show differences in the hippocampus compared to the ADNC0/LATENC0 subgroup ($p > 0.1$) (Figure 5).

When stratifying by ADNC severity, LATENC stage was independently associated with less hippocampal tissue when ADNC was

between 1 and 3 (for ADNC Level 1: partial $R^2 = 0.20$, $p = 6.9 \times 10^{-8}$, $n = 187$; for ADNC Level 2: partial $R^2 = 0.13$, $p = 5.8 \times 10^{-11}$, $n = 381$; for ADNC Level 3: partial $R^2 = 0.06$, $p = 0.006$, $n = 234$), but not when ADNC was 0 (partial $R^2 = 0.07$, $p = 0.12$, $n = 110$). Conversely, when stratifying by LATENC stage, ADNC severity showed no statistically significant association with hippocampal logJ (for LATENC Stage 0: partial $R^2 = 0.02$, $p = 0.08$, $n = 402$; for LATENC Stage 1: partial $R^2 = 0.02$, $p = 0.43$, $n = 159$; for LATENC Stage 2: partial $R^2 = 0.03$, $p = 0.53$, $n = 103$; for LATENC Stage 3: partial $R^2 = 0.03$, $p = 0.08$, $n = 248$). That

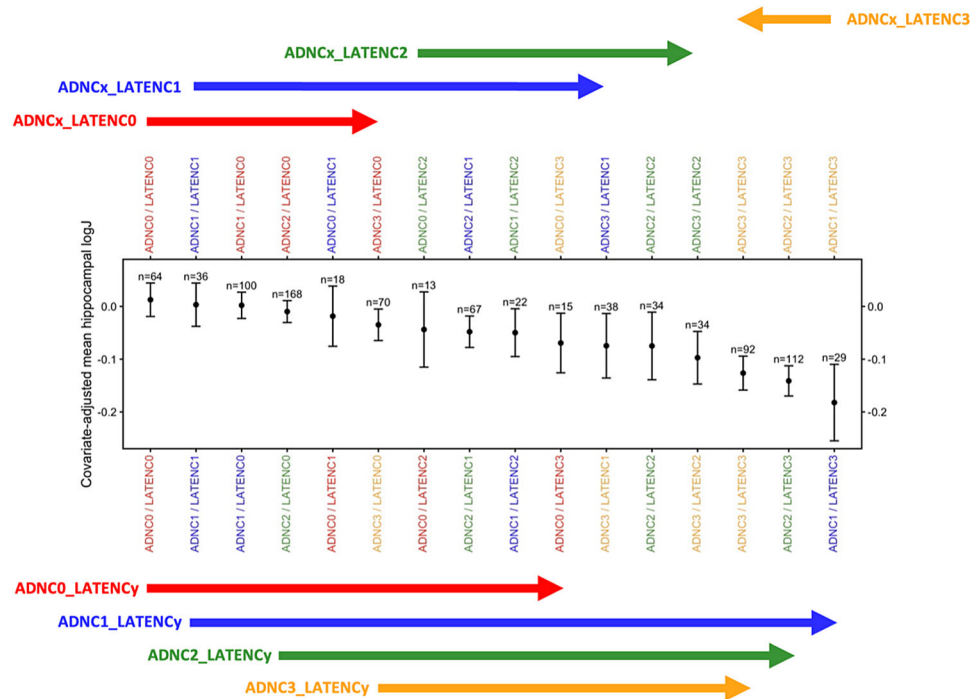


FIGURE 5 Covariate-adjusted estimated marginal means (EMMs) for hippocampal log-Jacobian determinant (logJ), ordered from higher to lower. Lower logJ values indicate less tissue relative to the template. Error bars represent 95% confidence intervals; subgroup sample sizes (*n*) are indicated above each point. Axis labels are color-coded by neuropathological stage (0 = red, 1 = blue, 2 = green, 3 = orange), where the bottom x-axis denotes Alzheimer's disease neuropathologic change (ADNC) severity level and the top x-axis denotes limbic-predominant age-related transactive response DNA-binding protein 43 kDa (TDP-43) encephalopathy (LATENC) stage. The arrows below the plot depict the range of differences across LATENC stages 0–3 within a given ADNC severity level (e.g., ADNC = 0 with LATENC ranging from 0 to 3). The arrows above the plot depict the range of differences across ADNC severity levels 0–3 within a given LATENC stage (e.g., LATENC = 0 with ADNC ranging from 0 to 3). For any given ADNC severity level (arrows below), more severe LATENC stage resulted in a large shift towards lower logJ values. In contrast, for any given LATENC stage (arrows above), more severe ADNC level resulted in a smaller shift toward lower logJ values.

is, LATENC stage explains more of the variation in hippocampal size than ADNC severity, independent of the severity of comorbid ADNC or comorbid LATENC, respectively.

4 | DISCUSSION

The present study combined ex vivo MRI DBM and detailed neuropathologic examination in a large number of community-based older adults and demonstrated similarities and differences in brain morphometry patterns associated with ADNC, LATENC, and mixed ADNC and LATENC, controlling for other neurodegenerative and vascular pathologies. The AD+LATE- and AD-LATE+ patterns had substantial overlap and showed less tissue, mainly in the medial temporal lobe, compared to AD-LATE- individuals. When both ADNC and LATENC were present, the brain morphometry pattern demonstrated less tissue across more regions than the AD+LATE- and AD-LATE+ patterns. When comparing the AD-LATE+ and AD+LATE- groups, the former showed smaller anterior hippocampi than the latter. When repeating the analysis in older adults without dementia, the above patterns demonstrated statistical significance in fewer and smaller regions. Furthermore, the difference in hippocampal size between more severe

versus less severe LATENC stages, for any severity level of comorbid ADNC, was more pronounced than the difference in hippocampal size between more severe versus less severe ADNC, for any severity level of comorbid LATENC. The above findings suggest deleterious effects of ADNC and LATENC on the brain, additive degeneration when both pathologies are present, and a stronger impact of LATENC compared to ADNC on the size of the hippocampus, especially on the anterior portion of the hippocampus. This enhanced understanding of the similarities and differences in brain morphometry patterns of ADNC and LATENC may have important implications for diagnosis of these pathologies in older adults.

Compared to AD-LATE- older adults, both AD+LATE- (Figure 1) and AD-LATE+ (Figure 2) individuals exhibited less tissue in medial temporal lobe structures, with largely overlapping morphometry patterns. These findings align with prior studies in ADNC and LATENC reporting smaller volumes,^{56,57} cortical thinning,^{14,17} reduced R2 relaxation rates,²⁸ lower diffusion anisotropy,^{33,58} compromised white matter connectivity, and other anomalies in the medial temporal lobe.²⁹ The brain regions included in the overlapping patterns are critical to memory, consistent with the amnesic clinical syndromes commonly observed in both diseases.^{9,59} Our analysis in individuals without dementia further revealed atrophy in the medial temporal

lobe in the AD-LATE+ group, suggesting that LATENC is associated with medial temporal degeneration even before the onset of substantial cognitive impairment, underscoring its role as an early and independent contributor to neurodegeneration.

The AD+LATE+ group exhibited less tissue across more regions than those involved in the AD+LATE- and AD-LATE+ patterns. These findings align with previous reports showing accelerated cognitive impairment and greater neuronal loss compared to either pathology alone.^{8,10,13,20-22,24,25,30,60} Individuals with mixed ADNC and LATENC show twice the rate of cognitive decline and memory impairment compared to those with only one of these pathologies.²⁴ The extent of the morphometric pattern observed in the AD+LATE+ group is consistent with a potential synergistic effect of ADNC and LATENC, resulting in more severe and widespread brain atrophy than seen in either pathology alone. This potential synergy may be driven by molecular interactions between TDP-43 and tau proteinopathies, as postmortem studies have reported their co-localization.^{4-6,26,61} Given the common comorbidity of LATENC and ADNC in older adults, understanding their combined impact is important.^{62,63} Mixed pathologies may lower the threshold for cognitive dysfunction, exacerbate neurodegeneration, and limit the utility of biomarkers targeting amyloid or tau alone.^{8,25,26} When repeating the analysis in individuals without dementia, the AD+LATE+ group continued to exhibit less tissue in the medial temporal lobe, indicating that the potential synergistic effects of ADNC and LATENC precede substantial cognitive impairment. Of interest, research in older adults without dementia has shown that less tissue in the derived AD+LATE+ pattern is associated with lower baseline cognitive performance and accelerated cognitive decline.⁶⁴

Compared to the AD+LATE- pattern of the present study, the AD+LATE+ pattern more closely resembled the known ADNC morphometric signature^{31,65-67} as it included temporal, frontal, and parietal regions, whereas the AD+LATE- pattern included mainly temporal and limited frontal regions. This limited extent of the AD+LATE- pattern may be related to the analytical approach employed here, which used categorical instead of continuous variables to compare the AD+LATE- subgroup to the AD-LATE- subgroup, and did not consider many individuals with ADNC that were assigned to the AD+LATE+ subgroup. Alternatively, because ADNC and LATENC often coexist and several previous studies on the morphometric signature of ADNC did not control for LATENC due to the lack of an appropriate marker or pathology data, it may be that part of what is known as the ADNC morphometric signature is actually the result of mixed ADNC and LATENC seen in the AD+LATE+ pattern.

Despite the substantial overlap of the AD+LATE- and AD-LATE+ patterns, our analysis revealed less tissue in the anterior hippocampus and adjacent portions of the entorhinal and parahippocampal cortices in the AD-LATE+ group compared to the AD+LATE- group, controlling for other pathologies and demographics. This finding is consistent with an anterior-to-posterior gradient of medial temporal lobe atrophy described previously in association with LATENC.¹⁴ Another study also found that LATENC burden was strongly linked to atrophy in anterior medial temporal lobe structures, particularly the anterior hippocampus and entorhinal cortex, whereas tau pathology showed relatively

greater involvement of the posterior hippocampus.¹⁵ Complementary work showed that greater LATENC burden is associated with reduced fractional anisotropy in white matter pathways connecting the anterior hippocampus to adjacent regions, particularly to the parahippocampal and entorhinal cortices.³³

Our analysis focusing exclusively on the hippocampus showed that the effect on hippocampal size was more pronounced with higher severity of LATENC, independent of the severity of mixed ADNC. This is in line with recent studies showing that hippocampal size for AD+LATE+ older adults is similar to that for AD-LATE+ individuals.^{30,32} Furthermore, longitudinal studies have demonstrated accelerated hippocampal degeneration attributable to LATENC, independently of ADNC presence.²³ Altogether, the above indicate that LATENC has a dominant role in hippocampal neurodegeneration. The finding that subgroups with early stages of LATENC and ADNC did not show significant differences in the hippocampus compared to the subgroup with no ADNC or LATENC should be interpreted carefully. The analysis in the hippocampus divided the participants into 16 subgroups thereby reducing the sample size for individual comparisons, and limiting the power to detect small effects. In addition, although DBM is voxel-wise, the analysis summarized the logJ across the whole hippocampus using the median value, thereby diluting localized effects of early ADNC or LATENC.

This study has several important strengths as well as a few limitations. First, the large number of community-based older adults ($N = 912$) with MRI and detailed pathology data provided excellent statistical power to detect subtle differences in morphometric patterns and enhanced generalizability, and allowed us to control for other neurodegenerative and vascular pathologies in the analyses. Second, the use of ex vivo MRI provided images free of motion artifacts, allowed imaging of the brain regardless of participants' ability to tolerate research in vivo MRI and therefore enabled inclusion of older adults with the full range of pathology severity and associated clinical symptoms or lack thereof, and eliminated the possibility of additional pathology developing after MRI. One potential limitation of the present study may be that ex vivo MRI and neuropathologic examination were conducted in a single hemisphere for each participant. However, the study used roughly equal numbers of left and right hemispheres, and there is no systematic hemispheric asymmetry of age-related neuropathologic burden. Another limitation is that due to the observational nature of this study one cannot draw conclusions about causation.

In summary, our study combined ex vivo MRI DBM and detailed neuropathologic examination in a large number of community-based older adults and demonstrated similarities and differences in brain morphometry patterns associated with ADNC, LATENC, and mixed ADNC and LATENC, controlling for other neurodegenerative and vascular pathologies. Our findings suggest harmful effects of ADNC and LATENC on the brain, additive degeneration when both pathologies are present, and a stronger impact of LATENC compared to ADNC on the size of the hippocampus, especially on the size of the anterior portion of the hippocampus. Our findings pave the way for multiple promising directions in future research. First, investigations of the

pathophysiological mechanisms underlying the hypothesized synergistic effects of ADNC and LATENC are warranted. Second, it is important to extend our understanding of the processes that make the size of the hippocampus, especially of the anterior hippocampus, more sensitive to LATENC than to ADNC severity. Third, the pathology-grounded anatomic patterns identified here may be used as features in machine learning models that may lead to the development of fully automated MRI-based markers of ADNC or LATENC.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

All participants provided written informed consent. Those participants that came to autopsy had also signed an anatomic gift act agreeing to brain donation at the time of death.

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SUPPORTING INFORMATION

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